Catalytic–Wall Taylor–Couette Reactor: What the Simulation Computes and Why Modulation Helps

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Abstract

This note documents the catalytic–wall mass-transfer problem solved in the Taylor–Couette reactor (TCR) simulation, a partial recreation of López-Guajardo et al., *Chem. Eng. J.* **489** (2024) 151174. It defines the governing equations, the boundary conditions that model the catalytic wall, exactly how *conversion* is computed from the flow field, the physical quantities that dictate it, and the mechanism by which unsteady (modulated) rotation enhances it. It closes with the reinforcement-learning reward formulation for the catalysis objective.

1 System

The reactor is a vertical annular gap between a rotating inner cylinder (radius $r_\mathrm{i} = 25.4\,\mathrm{mm}$) and a fixed outer cylinder (radius $r_\mathrm{o} = 31.75\,\mathrm{mm}$); gap width $d = r_\mathrm{o} - r_\mathrm{i} = 6.35\,\mathrm{mm}$, height $H = 190.5\,\mathrm{mm}$, aspect ratio $\Gamma = H/d = 30$. Rotation of the inner cylinder above a critical speed drives axisymmetric Taylor vortices that provide cross-gap transport. A reactant species of concentration c is fed at the top inlet and consumed at the *outer* (catalytic) wall.

2 Governing equations

The carrier fluid is an incompressible, Newtonian silicone oil ($\rho = 930\,\mathrm{kg/m^3}$, $\nu = 1.0753e{-}5\,\mathrm{m^2/s}$). The velocity field $\mathbf{u}$ and pressure p satisfy continuity and the Navier–Stokes equations,

$$\nabla \cdot \mathbf{u} = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\,\nabla^2 \mathbf{u}. \tag{2}$$

The reactant is transported as a scalar by advection and molecular diffusion,

$$\boxed{\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = D\,\nabla^2 c} \tag{3}$$

with diffusivity D. There is *no* volumetric reaction term: the chemistry lives entirely in the wall boundary condition below. The inner cylinder rotates at prescribed angular speed $\omega(t)$, which is the control input (constant ω_b, a modulated square wave, or a learned policy).

Boundary	Velocity	Species c
Inner wall (rotating)	$u_\theta = r_\mathrm{i}\,\omega(t)$, no-slip	$\partial c/\partial n = 0$ (inert)
Outer wall (catalytic)	no-slip, fixed	$c = 0$ (Dirichlet sink)
Inlet (top)	axial $u_\mathrm{ax} = Q/A$	$c = c_0$ (feed)
Outlet (bottom)	outflow	$\partial c/\partial n = 0$

Table 1: Boundary conditions. The catalytic wall is the key modelling choice.

3 Boundary conditions: modelling the catalytic wall

The catalytic wall is represented as a **Dirichlet sink** $c = 0$. Physically this is the limit of an infinitely fast surface reaction (Damköhler number $\mathrm{Da} \to \infty$): any reactant that reaches the wall is consumed instantly, so the surface concentration is pinned at zero and the process becomes *mass-transfer limited*—throttled by how fast species can be *delivered* to the wall, not by reaction kinetics. This is the central simplification that makes the species problem tractable while retaining the physics that modulation acts on.

4 How conversion is computed

4.1 Definition

Conversion is the fraction of fed reactant consumed between inlet and outlet. By mass balance on the reactant stream,

$$X \;=\; 1 - \frac{c_\mathrm{cup}}{c_0}, \qquad c_\mathrm{cup} \;=\; \frac{\displaystyle\int_\mathrm{out} c\,(\mathbf{u}\cdot\mathbf{n})\,dA}{\displaystyle\int_\mathrm{out} (\mathbf{u}\cdot\mathbf{n})\,dA}, \tag{4}$$

where c_cup is the *cup-mixing* (flux-weighted) average outlet concentration. The flux weighting is essential: conversion concerns how much reactant actually *leaves in the stream*, so each location is weighted by its volumetric flux $\mathbf{u}\cdot\mathbf{n}\,dA$, not merely its area. In the code, (4) is evaluated over the discrete outlet faces i as

$$c_\mathrm{cup} = \frac{\sum_i c_i\,|u_{z,i}|\,A_i}{\sum_i |u_{z,i}|\,A_i}, \qquad X = 1 - c_\mathrm{cup} \quad (c_0 = 1). \tag{5}$$

4.2 Equivalent wall-flux view

At steady state the reactant consumed at the catalytic wall equals the reactant removed from the stream:

$$\underbrace{\int_\mathrm{wall}\Big(-D\,\frac{\partial c}{\partial n}\Big)\,dA}_{\text{wall flux } N} \;=\; Q\,c_0\,X, \tag{6}$$

where Q is the volumetric flow rate. The simulation logs both X (`conv`) and N (`wallFlux`); (6) is the consistency check between them.

5 What dictates conversion

Three quantities set X:

1. **Residence time** $\tau = V/Q$: longer time in the reactor gives the reactant more opportunity to reach the wall, raising X.

2. **Wall mass-transfer rate**, expressed by the mass-transfer coefficient k or Sherwood number $\mathrm{Sh} = k\,d/D$. This is the *only* quantity the rotation controls, and the lever modulation pulls.

3. **Diffusivity** D.

For a wall-reaction reactor these combine, to leading order, as

$$X \;\approx\; 1 - \exp\!\left(-\frac{k\,A_{\mathrm{wall}}}{Q}\right), \tag{7}$$

so conversion increases monotonically with the mass-transfer coefficient k.

The boundary-layer mechanism. Near the catalytic wall the concentration falls from its bulk value c_{b} to 0 across a thin *concentration boundary layer* of thickness δ_c. The local diffusive flux is

$$j \;=\; -D\,\frac{\partial c}{\partial n}\bigg|_{\mathrm{wall}} \;\sim\; \frac{D\,c_{\mathrm{b}}}{\delta_c}, \qquad \Longrightarrow \qquad k \sim \frac{D}{\delta_c}. \tag{8}$$

In *steady* flow δ_c settles to a fixed thickness set by the balance of convection and diffusion (Lévêque scaling $\delta_c \sim d\,\mathrm{Sc}^{-1/3}$, with $\mathrm{Sc} = \nu/D$). **This is why a thinner boundary layer means more flux, hence higher conversion.**

6 Why modulation enhances conversion

A constant rotation rate produces a steady vortex field and therefore a steady, fixed δ_c. *Modulating* $\omega(t)$ makes the near-wall shear unsteady: each high-speed burst sweeps the depleted fluid off the wall and replaces it with fresh bulk fluid, momentarily collapsing δ_c. Because $k \sim D/\delta_c$ and flux is highest when δ_c is smallest, the *time-averaged* mass-transfer coefficient satisfies

$$\big\langle k \big\rangle_{\mathrm{modulated}} \;>\; k_{\mathrm{constant}}, \tag{9}$$

even at the same mean rotation rate, and by (7) conversion rises. This is the effect the paper reports and the reason an *unsteady* actuation can beat a steady one—a benefit that a bulk mixing-uniformity metric (which saturates) does not reveal, but the wall-transport/conversion metric does.

7 Modulation waveform

The modulated cases use a non-negative square wave of mean ω_b, duty cycle D, and period T: the inner cylinder spins at $\omega_{\mathrm{peak}} = \omega_b/D$ for a fraction D of each period and is idle ($\omega = 0$) otherwise, so the time mean is preserved,

$$\langle \omega \rangle = D\,\omega_{\mathrm{peak}} + (1 - D)\cdot 0 = \omega_b. \tag{10}$$

The reference case is $\omega_b = 500\,\mathrm{rpm}$, $D = 0.2$, $T = 30\,\mathrm{s}$, giving $\omega_{\mathrm{peak}} = 2500\,\mathrm{rpm}$.

Group	Definition	Role
Rotational Reynolds	$\mathrm{Re}_{\mathrm{rot}} = \omega r_{\mathrm{i}} d/\nu$	vortex regime
Péclet	$\mathrm{Pe} = Ud/D$	advection vs. diffusion
Schmidt	$\mathrm{Sc} = \nu/D$	momentum vs. species diffusion
Sherwood	$\mathrm{Sh} = kd/D$	dimensionless wall transfer

8 Dimensionless groups and simulation simplification

The paper operates at $\mathrm{Sc} \sim 10^5$ ($D \sim 1e-10\,\mathrm{m^2/s}$), for which δ_c is extremely thin and requires a graded near-wall mesh. The present simulation uses a moderate $D = 1e-8\,\mathrm{m^2/s}$ ($\mathrm{Sc} \sim 10^3$, $\mathrm{Pe} \sim 10^6$): still strongly mass-transfer limited, but with a boundary layer resolvable on the current uniform radial mesh. The *trend* (modulation > constant) is preserved; absolute conversion is higher than the paper's because of the larger D. Recovering the paper's Sc is a meshing follow-up.

9 Reinforcement-learning reward

For the *mixing* objective the agent minimised a segregation index $I_{\mathrm{mix}} \in [0,1]$ (lower = better mixed) and energy E:

$$R_{\mathrm{mix}} = -\alpha\, I_{\mathrm{mix}} - \beta\, E. \tag{11}$$

For the *catalysis* objective the agent should *maximise* conversion $X \in [0,1]$. Two equivalent forms:

$$R_{\mathrm{cat}} = +\alpha\, X - \beta\, E \qquad \text{(maximise conversion), or} \tag{12}$$

$$R_{\mathrm{cat}} = -\alpha\,(1-X) - \beta\, E = -\alpha\, c_{\mathrm{cup}} - \beta\, E \qquad \text{(minimise unconverted fraction).} \tag{13}$$

The second form is structurally identical to R_{mix}: replace the segregation index I_{mix} with the unconverted fraction $(1-X) = c_{\mathrm{cup}}$, which the simulation already logs.

The weight β is the experiment. With E normalised by a reference per-step energy E_{max}, a high-speed modulation step has $E/E_{\mathrm{max}} \gg 1$, so the energy penalty is large precisely when the agent modulates. Consequently:

- β too large $\Rightarrow$ the agent avoids modulation and collapses to a low constant speed;
- β too small $\Rightarrow$ the agent modulates maximally and merely rediscovers the square wave;
- intermediate β $\Rightarrow$ the agent finds a favourable point on the conversion–energy trade-off.

Thus β (with E_{max} recalibrated so a modulation-scale step gives $E/E_{\mathrm{max}} \sim 1$) parameterises the Pareto front between conversion and energy; sweeping it traces that front and is itself a reportable result.